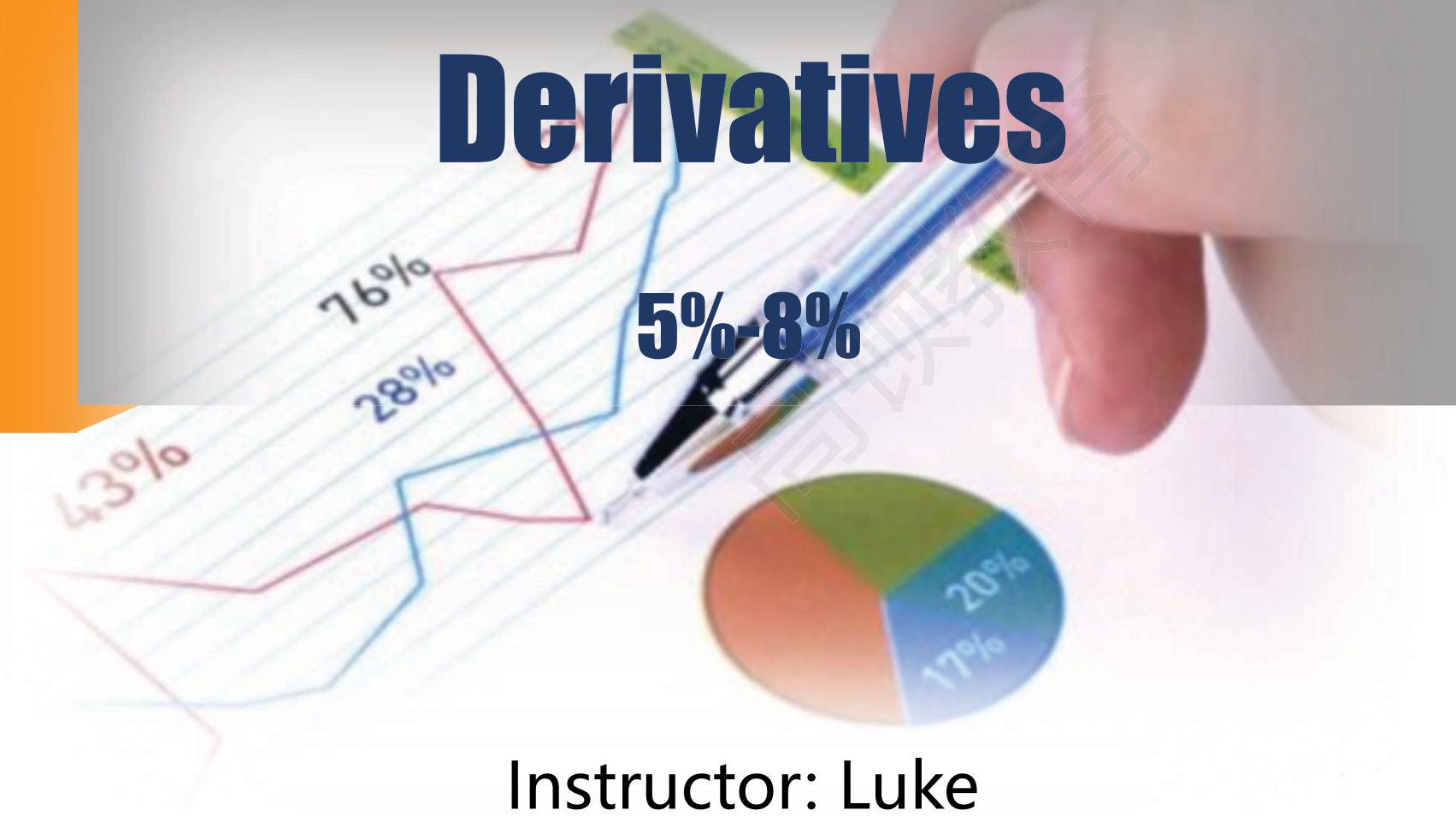
高顿教育
GOLDEN EDUCATION

Derivatives

5%-8%



Instructor: Luke

Derivative Instrument and Derivative Market Features

Tasks:

- Define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics;
- Describe the basic features of derivative markets, and contrast over-the-counter and exchange-traded derivative markets.

Derivative Instrument

Definition of derivative

- A derivative is a financial instrument that derives its performance from the performance of an underlying asset.
  - ✓ Financial asset: equity, fixed-income security, currency;
  - ✓ Physical asset: Soft commodity, Hard commodity;
  - ✓ Other: interest rate, credit, weather, longevity, other derivatives, etc.
- Derivative usually transform (not simply pass through) the performance of the underlying asset before paying it out in the derivatives transaction.

Derivative Instrument

Definition of derivative (Cont.)

- A derivative contract is a legal agreement between counterparties with a specific maturity.
  - ✓ The long: buyer, holder
  - ✓ The short: seller, writer

高顿教育

Derivative Instrument

Forward commitment

- Contracts entered into at one point in time that require both parties to engage in a transaction at a later point in time (the expiration) on terms agreed upon at the start.
  - ✓ Forward, futures, and swap

Contingent claim

- Derivatives in which the outcome or payoff is dependent on the outcome or payoff of an underlying asset.
  - ✓ Option

Derivative Market Features

Exchange-traded derivative (ETD) markets

- Standardized
- Regulated
- Transparent
- No default risk – guaranteed by clearinghouse

Over-the-counter (OTC) markets

- Customized
- Less regulated
- Less transparent
- Default risk/counterparty risk

Derivative Market Features

Exchange-traded markets

- Futures
- Some options

Over-the-counter (OTC) markets

- Forward
- Swap
- Some options

高顿教育

Derivative Market Features

Derivative markets

- Derivative markets expand the set of opportunities available to market participants.
  - ✓ Benefit from a decline in the value of the underlying.
  - ✓ Portfolio diversification.
  - ✓ Offset the financial market exposure. (Hedge)
  - ✓ Create large exposures with a relatively small cash outlay.
  - ✓ Derivatives typically have lower transaction costs and are often more liquid.

Summary

- **Importance:** ☆☆
- **Content:**
  - ✓ Derivatives: underlying, transform;
  - ✓ Forward, futures, swap, option;
  - ✓ Forward commitment Vs. Contingent claim;
  - ✓ Exchange-traded Vs. Over-the-counter.
- **Exam tips:**
  - ✓ 常考点：Exchange-traded和OTC market的特征对比。

高顿教育

Foward

Tasks:

- Define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics;
- Contrast forward commitments with contingent claims.

Forward

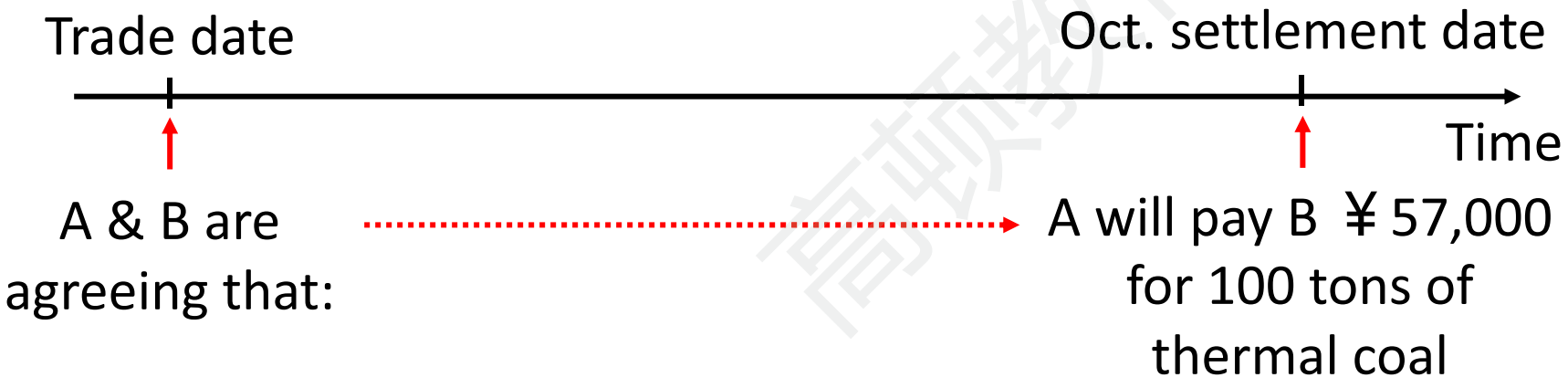
Forward

- An over-the-counter derivative contract in which two parties agree that one party, the buyer, will purchase an underlying asset from the other party, the seller, at a later date at a fixed price they agree on when the contract is signed.
- In addition to the (forward) price, the two parties also agree on several other matters, such as the identity and the quantity of the underlying.

Forward

Example:

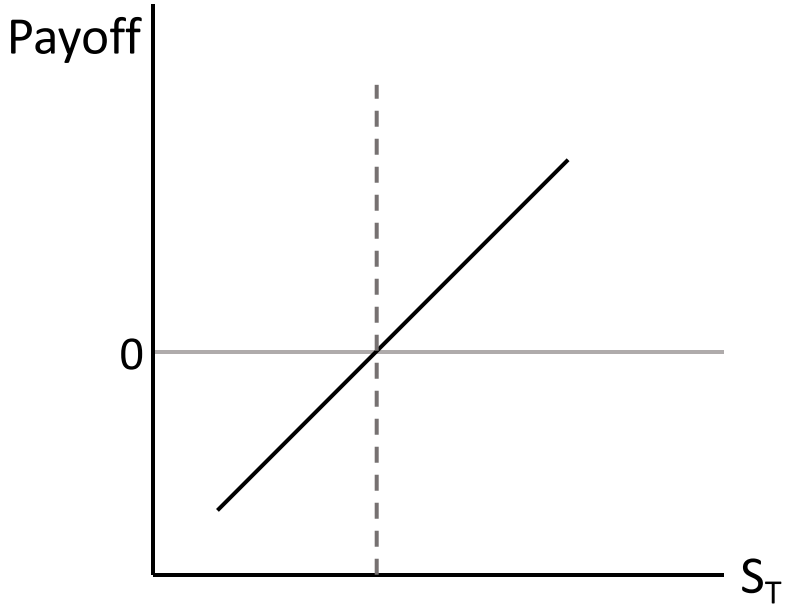
- Suppose that A contracts to buy 100 tons of thermal coal from B on 15th Oct. at ￥570 per ton.



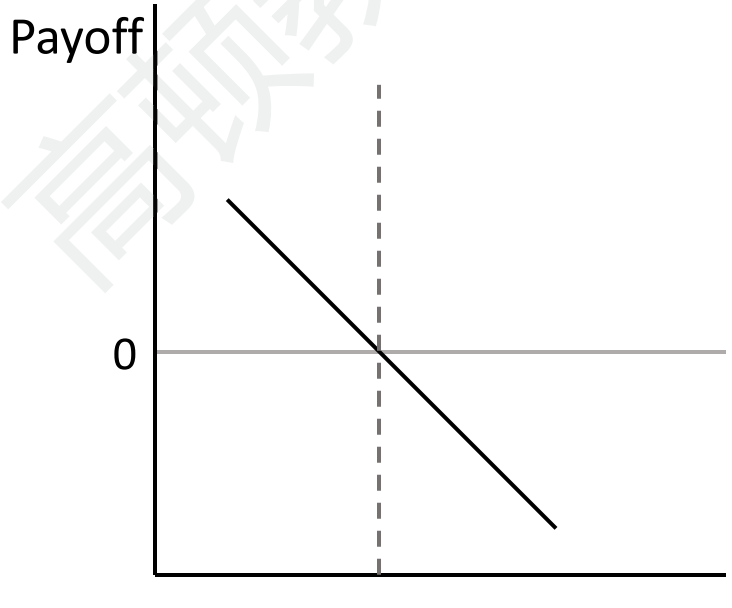
- ✓ A, who has agreed to buy, has the long forward position and is referred to as the long.
- ✓ B has the short forward position, and is called the short.

Forward (Cont.)

- The long hopes the price of the underlying will rise above the forward price, $F_0(T)$, whereas the short hopes the price of the underlying will fall below the forward price.



Payoff of the long $= S_T - F_0(T)$



Payoff of the short $= F_0(T) - S_T$

Forward (Cont.)

- Forward contracts can be settled in two ways:
  - ✓ Delivery of the underlying asset
  - ✓ Exchange of cash: non-deliverable forwards (NDFs), cash-settled forwards, or contracts for differences
- Forward contracts can be structured to create a perfect hedge, providing an assurance that the underlying asset can be bought or sold at a price known when the contract is initiated.

Forward (Cont.)

- Dealer: a dealer creates a derivative contract and will quote a price to take a long or short position;
  - ✓ A.k.a market maker or financial intermediary.
- End user: an end user is typically a corporation or institution seeking to transfer an existing risk.

Forward

Equity index forward

- An equity index contract payoff is the notional amount times the percentage difference between the forward index and the actual index value at settlement.

Example:

- ✓ 90-day S&P 100 forward contract
- ✓ Forward contract price = 500; In 90 days index is at 550
- ✓ Notional amount = $10 million
- ✓ What is the payment at settlement and which party receives it?

Equity index forward (Cont.)

- Solution:
  - ✓ Long receives (550 – 500) / 500 × $10 million = $1 million at settlement, paid by the short.

高顿教育

Summary

- **Importance:** ☆☆
- **Content:**
  - ✓ Basics of forwards;
  - ✓ Settlement of forwards;
  - ✓ Participants of forward market;
  - ✓ Equity index forward.
- **Exam tips:**
  - ✓ 常考点：Forward的的基本特征（与其他衍生品对比）。

    Forward payoff的计算

高顿教育 GOLDEN EDUCATION

Futures

Tasks:

- Define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics;
- Contrast forward commitments with contingent claims.

Futures

Futures

- Futures contracts are specialized forward contracts that have been standardized and trade on a futures exchange.
  - ✓ Futures contracts have specific underlying assets, times to expiration, delivery and settlement conditions, and quantities.
  - ✓ The exchange offers a facility in the form of a physical location and/or an electronic system as well as liquidity provided by authorized market makers.

Futures

Futures (Cont.)

- **Futures price**: the agree-upon price, like forward price.
- **Price limit**: a provision limiting price changes, establish a band relative to the previous day's settlement price, within which all trades must occur.
  - ✓ **Limit up/limit down**: trading stops if trading price is above/below the upper/lower band.
  - ✓ **Locked limit**: when market hits limits and trading stops.

Futures (Cont.)

- Mark-to-market: daily settlement of gains and losses to margin account according to the settlement price.
  - ✓ **Settlement price**: an average of the final futures trades of the day.

Futures (Cont.)

- **Initial margin:** both parties deposit a required minimum sum of money when the contract is initiated.
- **Maintenance margin:** the amount of money that must maintain in the margin account after the trade is initiated.
  - ✓ Always significantly lower than the initial margin.
- **Margin call:** a request to deposit enough funds to bring the margin account balance up to the initial margin.
  - ✓ The amount required is referred to as **variation margin**.

Futures (Cont.)

- There are significant differences between futures margin accounts and equity margin accounts:
  - ✓ A futures margin is simply an amount of money put into an account that covers possible future losses, and there is no formal loan created as in equity markets.
  - ✓ For equity margin account, an investor deposits part of the cost of the stock and borrows the remainder at a rate of interest.

Futures (Cont.)

- **Offset/close-out:** a party re-enters the market at a later date but before expiration and engages in the opposite transaction.
- **Open interest:** the number of outstanding contracts at any given time.
  - ✓ Each contract counted in the open interest has a long and a corresponding short.

高顿教育 GOLDEN EDUCATION

Futures

Forward vs. Futures

- Forward contracts realize the full gain or loss at expiration, whereas futures contracts realize the gain or loss in parts on a day-to-day basis.
  - ✓ The time value of money makes these not equivalent, but the differences tend to be small.
  - ✓ In forward contract, with the entire payoff made at expiration, a loss by one party can be large enough to trigger a default.

Summary

- **Importance:** ☆☆
- **Content:**
  - ✓ Basics of fututres;
  - ✓ Margins;
  - ✓ Mark-to-market;
  - ✓ Open interest.
- **Exam tips:**
  - ✓ 常考点：Futures的基本特征（与其他衍生品对比）；
    每日结算、保证金制度。

高顿教育

Swap

Tasks:

- Define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics;
- Contrast forward commitments with contingent claims.

Swap

Swap

- An over-the-counter derivative contract in which two parties agree to exchange a series of cash flows whereby one party pays a variable series that will be determined by an underlying asset or rate and the other party pays either (1) a variable series determined by a different underlying asset or rate or (2) a fixed series.
  - ✓ A swap is a series of (off-market) forwards.

Swap (Cont.)

- Characteristics of swap contracts:
  - ✓ Custom instruments;
  - ✓ Not traded in any organized secondary market;
  - ✓ Largely unregulated;
  - ✓ Default risk is a concern;
  - ✓ Most participants are large institutions;
  - ✓ Private agreements;
  - ✓ Difficult to alter or terminate.

高顿教育 GOLDEN EDUCATION

Swap (Cont.)

- Interest rate swap
  - ✓ Fixed for floating (plain vanilla)
  - ✓ Floating for floating
- Equity swap
- Currency swap

高顿教育

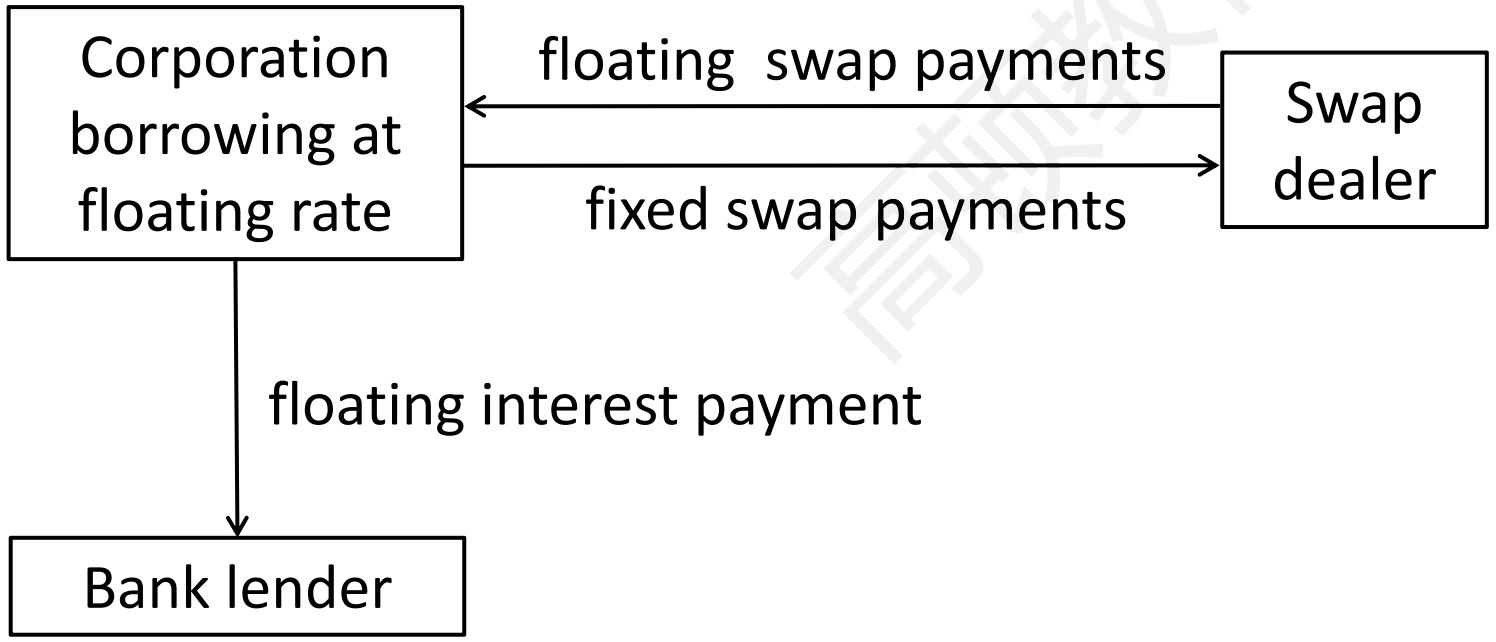
Swap

Plain Vanilla interest rate swap

- Notional amount is not exchanged at the beginning or end of the swap (both loans are in same currency and amount);
- On settlement dates, interest payments are netted;
  - ✓ Only the difference is paid by the party owing the greater amount.
- Floating rate payments are typically made in arrears.
  - ✓ Payment is made at end of period based on beginning-of-period market reference rate.

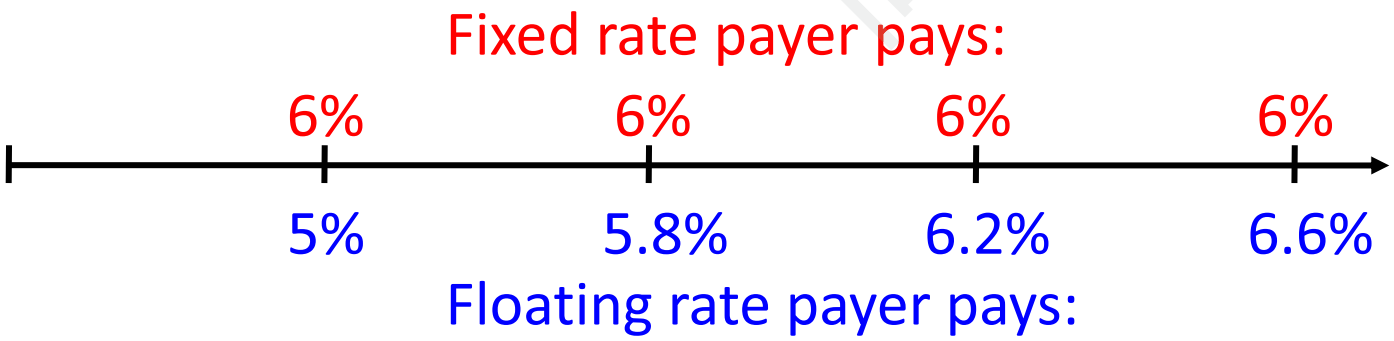
Plain Vanilla interest rate swap (Cont.)

- **Example**: using an interest swap to convert a floating rate loan to a fixed rate loan:



Plain Vanilla interest rate swap (Cont.)

- **Example**: a 2-year semiannual-pay plain vanilla interest rate swap on LIBOR for $10 million with a fixed rate of 6%. If LIBOR at initiation, in 0.5 year, in 1 year, and in 1.5 year are 5%, 5.8%, 6.2% and 6.6% respectively. Clarify the payments of the interest rate swap.



Swap

Solution:

- 1st payment: fixed-rate payer pays $50,000 net:
  - ✓ $(0.06 - \mathbf{0.05}) \times (180/360) \times (10 \text{ million}) = \$50{,}000$
- 2nd payment: fixed rate payer pays $10,000 net:
  - ✓ $(0.06 - \mathbf{0.058}) \times (180/360) \times (10 \text{ million}) = \$10{,}000$
- 3rd payment: floating rate payer pays $10,000 net:
  - ✓ $(0.06 - \mathbf{0.062}) \times (180/360) \times (10 \text{ million}) = -\$10{,}000$
- 4th payment: floating rate payer pays $30,000 net:
  - ✓ $(0.06 - \mathbf{0.066}) \times (180/360) \times (10 \text{ million}) = -\$30{,}000$

Summary

- **Importance:** ☆☆
- **Content:**
  - ✓ Basics of swaps;
  - ✓ Interest rate swap.
- **Exam tips:**
  - ✓ 常考点：Swaps的基本特征（与其他衍生品对比）；
    利率互换的机制。

高顿教育

Option and Credit Derivatives

Tasks:

- Define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics;
- Determine the value at expiration and profit from a long or a short position in a call or put option.
- Contrast forward commitments with contingent claims.

Option

Option

- A derivative contract in which one party, the buyer, pays a sum of money to the other party, the seller or writer, and receives the right to either buy or sell an underlying asset at a fixed price either on a specific expiration date or at any time prior to the expiration date.
  - ✓ An option is a right, but not an obligation.
  - ✓ Default in options is possible only from the short to the long.

Option

Option (Cont.)

- **Option premium** (c_0, p_0): payment to seller from buyer
- **Call option**: right to buy
- **Put option**: right to sell
- **Exercise price/strike price** (X): the fixed price at which the underlying asset can be purchased
- **American option**: exercisable at or prior to expiration
- **European option**: exercisable only at expiration

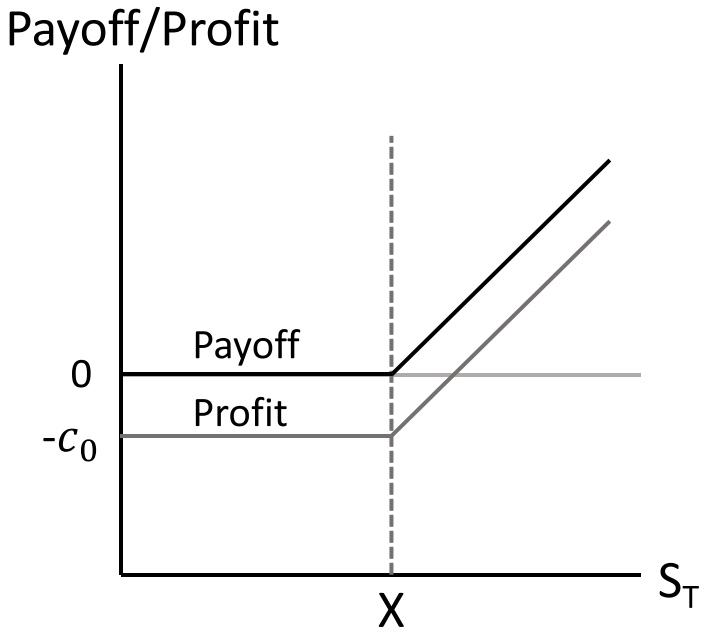
Option (Cont.)

- **Moneyness** (X: exercise price; S_t : spot price at time t)
  - ✓ In the money
    - Call: $S_t > X$
    - Put: $X > S_t$
  - ✓ At the money
    - $S_t = X$
  - ✓ Out of the money
    - Call: $S_t < X$
    - Put: $X < S_t$

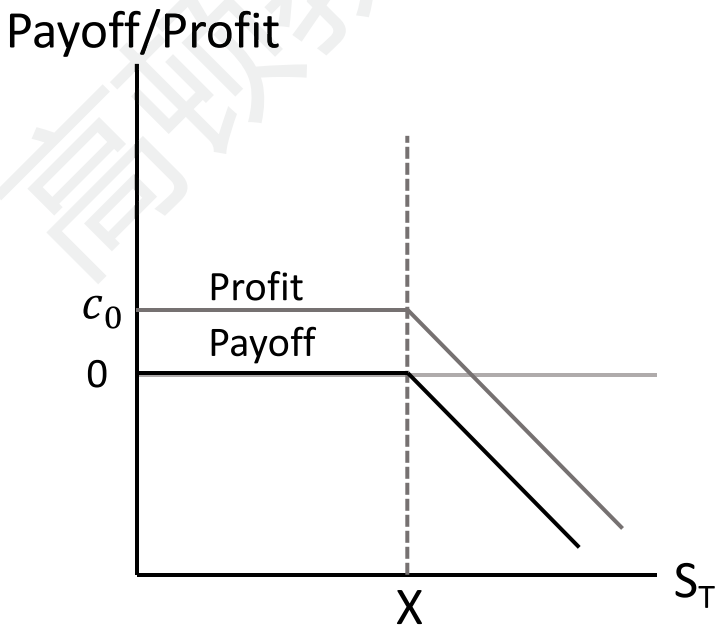
高顿教育

Option (Cont.)

- Payoff (c_T)and profit ($\prod$)from a call option at expiration
  - ✓ From buying: $c_T = \mathrm{Max}(0, S_T - X); \prod = \mathrm{Max}(0, S_T - X) - c_0$
  - ✓ From selling: $-c_T = -\mathrm{Max}(0, S_T - X); \prod = -\mathrm{Max}(0, S_T - X) + c_0$



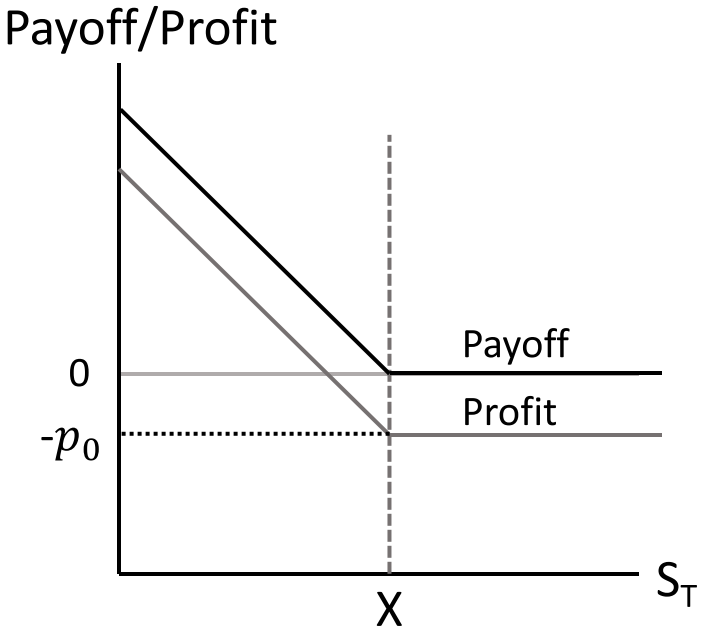
Payoff/Profit from buying



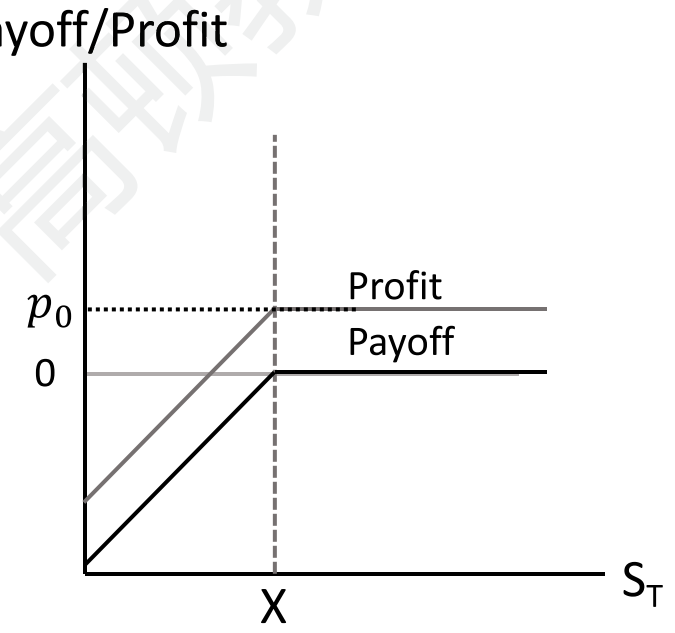
Payoff/Profit from selling

Option (Cont.)

- Payoff (p_T)and profit ($\prod$)from a put option at expiration
  - ✓ From buying: $p_T = \mathrm{Max}(0,\ X - S_T); \prod = \mathrm{Max}(0,\ X - S_T) - p_0$
  - ✓ From selling: $-p_T = -\mathrm{Max}(0,\ X - S_T); \prod = -\mathrm{Max}(0,\ X - S_T) + p_0$



Payoff/Profit from buying



Payoff/Profit from selling

Credit Derivatives

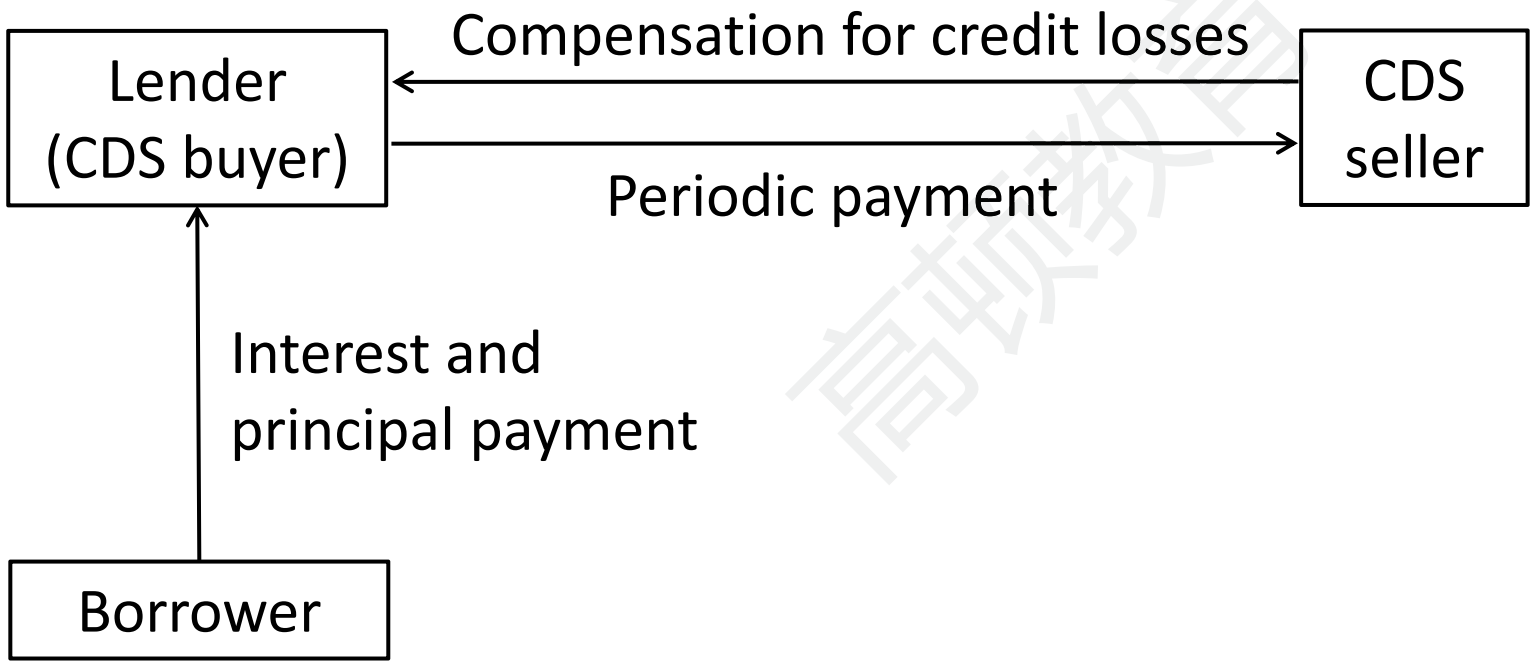
Credit default swap

- Derivative contracts between two parties, a credit protection buyer and a credit protection seller, in which the latter provides protection to the former against a specific credit loss.
  - ✓ Credit default swap (CDS): the buyer makes a series of cash payments to the seller and receives a promise of compensation for credit losses resulting from the default of a third party.
  - ✓ CDS is essentially an insurance contract against default.

Credit Derivatives

Credit default swap (Cont.)

- Example: using an CDS to hedge the credit risk of a loan:



Credit Derivatives

Other credit derivatives

- Total return swap
- Credit-linked note
- Credit spread option

高顿教育

Summary

- **Importance:** ☆☆☆
- **Content:**
  - ✓ Basics of options;
  - ✓ Moneyness;
  - ✓ Payoff and profit;
  - ✓ Credit default swap.
- **Exam tips:**
  - ✓ 常考点：Options的基本特征（与其他衍生品对比）；
    期权收益与利润的计算；
    信用违约互换机制定性考察。

高顿教育

Derivative Benefits, Risks, and Issuer and Investor Uses

Tasks:

- Describe benefits and risks of derivative instruments;
- Compare the use of derivatives among issuers and investors.

Derivative Benefits, Risks

Derivatives benefits

- Risk allocation, transfer, and management
- Information discovery
  - ✓ Price discovery
  - ✓ implied volatility
- Operational advantages
  - ✓ Lower transaction cost
  - ✓ Greater liquidity
  - ✓ Easy to go short
- Better market efficiency

高顿教育

Derivative Benefits, Risks

Derivative risks

- Speculation and gambling
- High leverage - too risky
- Complexity
  - ✓ Strucutured notes = debt instruments + derivatives
- Basis risk
- Liquidity risk
- Counterparty credit risk
- Destabilization and systemic risk

Issuer and Investor Uses

Issuer uses

- Issuers, investors, and financial intermediaries use derivative instruments to meet their financial objectives.
- Non-financial corporate issuers often hedge risks to their assets, liabilities, and earnings.
  - ✓ Cash flow hedges.
  - ✓ Fair value hedge.
  - ✓ Net investment hedges.

Issuer and Investor Uses

Issuer uses (Cont.)

- Any derivative purchased or sold must be marked to market through the income statement via earnings.
  - ✓ Unless it is embedded in an asset or liability or qualifies for hedge accounting.
- In order to qualify for this treatment, the features of a derivative must closely match those of the underlying transaction.
  - ✓ OTC derivatives are preferred.

Issuer and Investor Uses

Investor uses

- Investors use derivatives to replicate a cash market strategy, hedge a fund's value against adverse movements in underlyings, or modify or add exposures.
- Less focused than issuers on hedge accounting treatment.
  - ✓ Exchange-traded derivatives are more frequently used.

Summary

- **Importance:** ☆
- **Content:**
  - ✓ Derivatives benefits and risks;
  - ✓ Issuer and investor uses.
- **Exam tips:**
  - ✓ 常考点：无。

高顿教育

Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

Tasks:

- Explain how the concepts of arbitrage and replication are used in pricing derivatives.
- Explain the difference between the spot and expected future price of an underlying and the cost of carry associated with holding the underlying asset.

Arbitrage

Arbitrage

- Law of one price: assets that produce identical future cash flows regardless of future events should have the same price.
- Arbitrage is a type of transaction undertaken when two assets or portfolios produce identical results but sell for different prices.
  - ✓ Riskless with zero investment.
  - ✓ Trader will exploit the arbitrage opportunity quickly (buy low and sell high), then make the prices converge.

Arbitrage (Cont.)

- A derivative-related arbitrage opportunity arises when an asset with a known future price does not trade at the present value (PV) of its future price.
  - ✓ The appropriate discount rate is the risk-free rate.
- No arbitrage pricing
  - ✓ $S_0 = S_T(1+r)^{-T} \quad \rightarrow \quad F_0(T) = S_0(1+r)^T$
    - S_0 is the spot price and S_T is the known future price.
  - ✓ $S_0 = S_T e^{-rT} \quad \rightarrow \quad F_0(T) = S_0 e^{rT}$
    - For continuous compounding

Replication

Replication

- A asset's cash flow stream may be recreated using a comination of other assets.
- An asset and a hedging position of derivative on the asset can be combined to produce a position equivalent to a risk-free asset.
  - ✓ Asset - Derivative = Risk-free asset
  - ✓ Asset - Risk-free asset = Derivative
  - ✓ Derivative + Risk-free asset = Asset

Cost of Carry

Carring costs and carring benefits

- Many assets have additional costs or benefits of ownership that must be reflected in the forward commitment price in order to prevent riskless arbitrage opportunities.
- No arbitrage pricing
  - ✓ $F_0(T) = [S_0 - PV_0(I) + PV_0(C)](1 + r)^T$
    - $PV_0(I)$ is the present value of carring benefits.
    - $PV_0(C)$ is the present value of carring costs.
  - ✓ $F_0(T) = S_0 e^{(r+c-i)T}$
    - For continuous compounding

Cost of Carry

Carring costs and carring benefits (Cont.)

- Opportunity cost (*r*)
  - ✓ Risk-free interest rate
- Other costs of ownership (*C, c*)
  - ✓ Storage, transportation, insurance, and spoilage costs.
- Benefits of ownership (*I, i*)
  - ✓ Dividend, interest, rent, convenience yield, etc.
    - Convenience yield: nonmonetary advantage of holding the asset.

Cost of Carry

Foward vs. Spot price

- Opportunity and Other Cost = Benefit
  - ✓ $F_0(T) = S_0$
- Opportunity and Other Cost > Benefit
  - ✓ $F_0(T) > S_0$ (Contango)
- Opportunity and Other Cost < Benefit
  - ✓ $F_0(T) < S_0$ (Backwardation)

高顿教育

Summary

- **Importance:** ☆☆☆
- **Content:**
  - ✓ Arbitrage & replication;
  - ✓ Cost of carry.
- **Exam tips:**
  - ✓ 是derivative定价的基础;
  - ✓ 常考点：套利机制的定性考察。
    持有成本和收益的影响。

高顿教育

Pricing and Valuation of Forward

Tasks:

- Explain how the value and price of a forward contract are determined at initiation, during the life of the contract, and at expiration;
- Explain how forward rates are determined for an underlying with a term structure and describe their uses.

Pricing and Valuation of Forward

Price of forward, futures, and swap

- The fixed price or rate at which the underlying will be purchased at a later date.
  - ✓ Generally may not change as the (expected) price of the underlying asset changes.

Value of forward, futures, and swap

- The difference of “with the position” from “without the position”.
  - ✓ May increase or decrease as the (expected) price of the underlying asset changes.

Pricing and Valuation of Forward

At initiation

- The forward contract at initiation has zero value.
  - ✓ Neither party to a forward transaction pays to enter the contract at initiation.
    - $V_0(T) = 0$
- The forward price must be set so its value at initiation is zero.
- $F_0(T) = [S_0 - PV_0(I) + PV_0(C)](1 + r)^T$
  - ✓ If no cash flows:
    - $F_0(T) = S_0(1 + r)^T$

Pricing and Valuation of Forward

During its life

- In the financial world, we generally define value as the value to the long position.
- At time t < T, the value of a forward contract is the spot price of the asset minus the present value of the forward price, and minus the net cost of carry.
- $V_t(T) = [S_t - PV_t(I) + PV_t(C)] - F_0(T)(1+r)^{-(T-t)}$
  - ✓ If no cash flows:
    - $V_t(T) = S_t - F_0(T)(1+r)^{-(T-t)}$

Pricing and Valuation of Forward

At expiration

- The value of a forward contract is the spot price of the underlying minus the forward price.
  - ✓ $V_T(T) = S_T - F_0(T)$

高顿教育

Pricing and Valuation of Forward

Benefits and costs of holding the underlying

- Benefits will decrease forward price at initiation, and decrease the value of forward contract during its life.
  - ✓ Including dividend, interest, rent, convenience yield, etc.
- Costs will increase forward price at initiation, and increase the value of forward contract during its life.
  - ✓ Including storage cost, maintenance cost, etc.

Pricing and Valuation of Forward

Definition of FRA

- Forward contracts in which the underlying is an interest rate (e.g. MRR) are called forward rate agreements, or FRAs.
  - ✓ Long position can be viewed as the obligation to take a (hypothetical) loan at the contract rate (i.e., borrow at the fixed rate); gains when reference rate increase.
  - ✓ Short position can be viewed as the obligation to make a (hypothetical) loan at the contract rate (i.e., lend at the contract rate); gains when reference rate decrease.

Pricing and Valuation of Forward

The uses of FRA

- Lock the interest rate or hedge the risk of borrowing or lending at some future date.
  - ✓ One party will pay the other party the difference (based on notional value) between the interest rate specified in the FRA and the market interest rate at contract settlement.

Pricing and Valuation of Forward

FRA settlement example

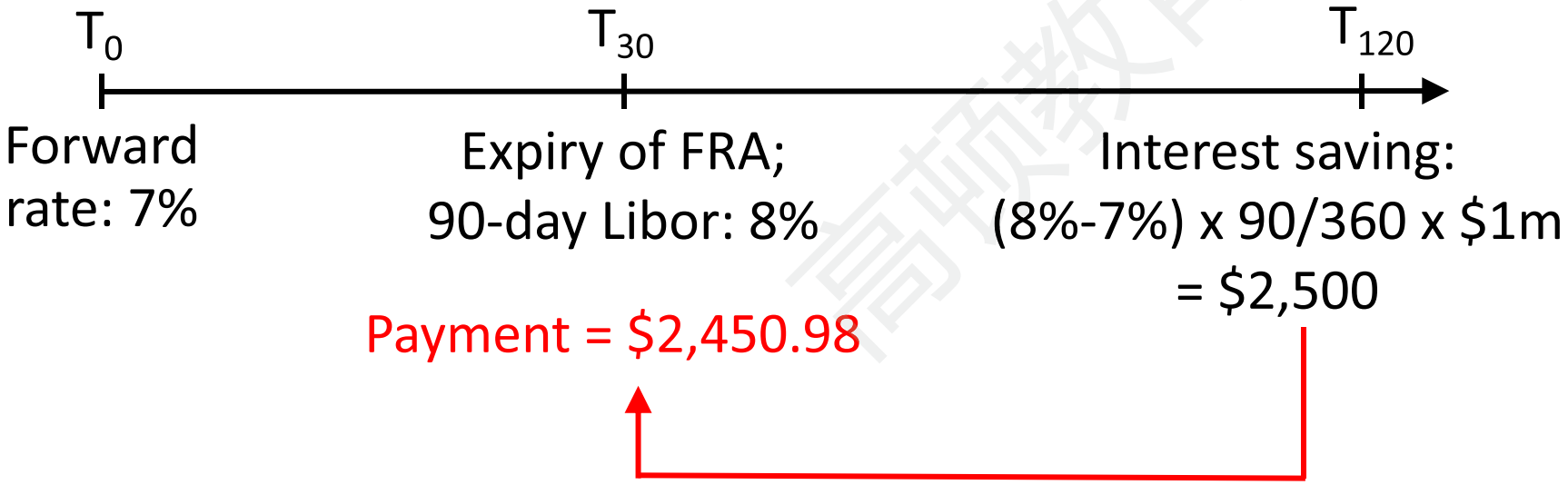
- Specification of 1 × 4 FRA:
  - ✓ Term = 30 days
  - ✓ Notional amount = $1 million
  - ✓ Underlying rate = 90-day LIBOR
  - ✓ Forward rate = 7%
- At t = 30 days, 90-day LIBOR = 8%, clarify the payment of this FRA.

高顿教育

Pricing and Valuation of Forward

FRA settlement example (Cont.)

- Underlying floating rate (market rate on T30) > fixed rate (contract rate) , so long position receives payment.



Pricing and Valuation of Forward

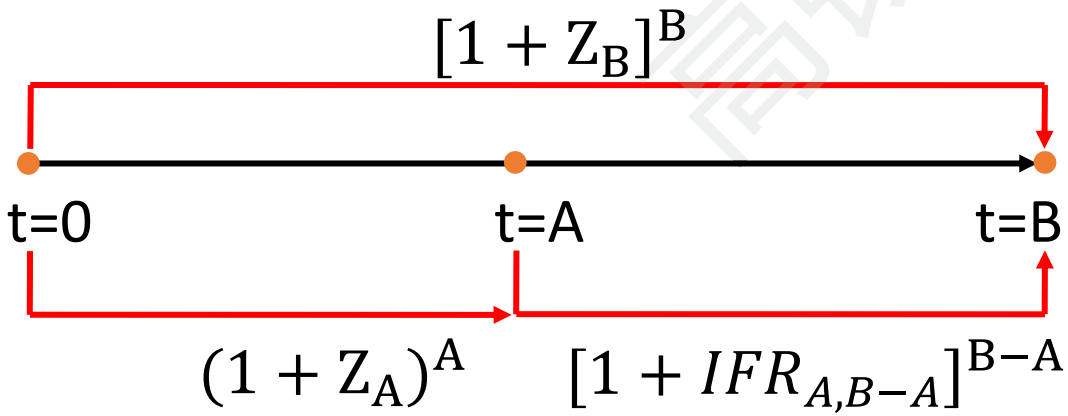
Spot rates and forward rates

- **Spot rate**: the rate of interest on a security that makes a single payment at a future point in time (a.k.a. or zero rate).
  - ✓ **Discount factor**: the present value of a currency unit on a future date.
    - $DF_i = \frac{1}{(1+Z_i)^i}$
- **Forward rate**: rate of interest set today for a single-payment security to be issued at a future date.
  - ✓ **Implied forward rate**: obtained form spot rates.

Pricing and Valuation of Forward

Forward rate models

- Forward rate models show how implied forward rates can be extrapolated from spot rates.
  - ✓ $[1 + \mathrm{Z_B}]^{\mathrm{B}} = (1 + \mathrm{Z_A})^{\mathrm{A}} \times [1 + IFR_{A,B-A}]^{\mathrm{B-A}}$



Summary

- **Importance:** ☆☆☆
- **Content:**
  - ✓ Forward price and value;
  - ✓ Forward rate agreement.
- **Exam tips:**
  - ✓ 常考点：Forward的price的计算；

    Cost和benefit对forward价格和价值的影响；

    Forward rate的计算。

高顿教育 GOLDEN EDUCATION

Pricing and Valuation of Forward

FX (foreign exchange) forward contract

- Spot foregin exchange rate ($S_{0,f/d}$)
  - ✓ f is the foreign (price) currency
  - ✓ d is the domestic (base) currency
- FX forward contract: sale of one currency and purchase of the other on a future date at a forward exchange rate ($F_{0,f/d}$).
  - ✓ Long (short) position means purchasing (selling) the domestic currency and selling (purchasing) the foreign currency.

Pricing and Valuation of Forward

No arbitrage pricing

- $F_{0,f/d}(T) = S_{0,f/d} \times e^{(r_f - r_d)T}$
  - ✓ r_f is the risk-free rate of the foreign currency.
  - ✓ r_d is the risk-free rate of the domestic currency.
- The spot versus forward FX price relationship is determined by the risk-free interest rate differential.

Interest rate differential	Forward vs. Spot
$r_f - r_d > 0$	$F_{0,f/d}(T) > S_{0,f/d}$
$r_f - r_d < 0$	$F_{0,f/d}(T) < S_{0,f/d}$
$r_f - r_d = 0$	$F_{0,f/d}(T) = S_{0,f/d}$

Pricing and Valuation of Forward

Valuation of FX forward

- At any given time t, the MTM value of the FX forward is the difference between the current spot FX price ($S_{t,f/d}$) and the present value of the forward price discounted by the current difference in risk-free rates ($r_f - r_d$) for the remaining period through maturity
  - ✓ $V_t(T) = S_{t,f/d} - F_{0,f/d}(T) \times e^{-(r_f - r_d)(T-t)}$

Pricing and Valuation of Forward

Example

- A Baywhite client has entered into a long six-month MXN/USD FX forward contract. The MXN/USD spot exchange rate at inception is 19.8248, the six-month MXN risk-free rate is 4.25%, and the six-month USD risk-free rate is 0.5%. Baywhite's market strategist predicts that the Mexican central bank will surprise the market with a 50 bp short-term rate cut at its upcoming meeting. How the client's existing FX forward contract will be impacted if this prediction is realized and other parameters remain unchanged?

Pricing and Valuation of Forward

Answer

- The original MXN/USD forward exchange rate at inception is equal to:
  - ✓ $F_{0,f/d}(T) = 19.8248 \times e^{(4.25\%-0.5\%)\times 0.5} = 20.20$
- A new forward exchange rate would be equal to:
  - ✓ $F_{0,f/d}(T) = 19.8248 \times e^{(3.75\%-0.5\%)\times 0.5} = 20.15$
- MTM loss is equal to 0.05
- $V_t(T) = 19.8248 - 20.20 \times e^{-(3.75\%-0.5\%)\times 0.5} = -0.05$

Pricing and Valuation of Futures

Tasks:

- Compare the value and price of forward and futures contracts;
- Explain why forward and futures prices differ.

Pricing and Valuation of Futures

Pricing of Futures

- Many of the pricing and valuation principles associated with forward commitments are common to both forward and futures contracts.
- $V_0(T) = 0$
- $f_0(\mathrm{T}) = [S_0 - PV_0(I) + PV_0(C)](1 + r)^T$
  - ✓ If no cash flows:
    - $f_0(\mathrm{T}) = \mathrm{S}_0(1 + \mathrm{r})^{\mathrm{T}}$

高顿教育

Pricing and Valuation of Futures

Futures vs. Forward

- Futures realize the gain or loss in parts on a day-to-day basis, while forwards realize the full gain or loss at expiration.
  - ✓ The forward contract price remains fixed until the contract matures.
  - ✓ Futures contract prices fluctuate daily based upon market changes.
  - ✓ The cumulative realized gain or loss is similar.
    - The time value of money makes these not equivalent.

Pricing and Valuation of Futures

Futures vs. Forward (Cont.)

- Futures price will be higher than forward price when interest rate and futures price are positively correlated, and will be lower when they are negatively correlated.
  - ✓ A positive correlation between interest rate and futures price means that (for a long position) daily settlement provides funds (excess margin) when rates are high and they can earn more interest, and daily settlement requires funds (margin call) when rates are low and they only ask for less interest.
  - ✓ This is so called **convexity bias**.

Pricing and Valuation of Futures

Interest rate futures

- Interest rate futures trade on a price basis.
  - ✓ $f_{A,B-A} = 100 - (100 \times MRR_{A,B-A})$
    - $f_{A,B-A}$ is the futures price for a market reference rate.
  - ✓ E.g. $f_{3m,3m} = 98$ means $MRR_{3m,3m} = 2\%$
- This (100 − yield) price convention results in an inverse price/yield relationship.
  - ✓ A long futures position gains when reference rate decrease.
  - ✓ A short futures position gains when reference rate increase.

Pricing and Valuation of Futures

Interest rate futures (Cont.)

- Futures contract basis point value (BPV)
  - ✓ $\text{BPV} = \text{Notional Principal} \times 0.01\% \times \text{Period}$
- Example: A futures contract for three-month MRR with a $1,000,000 notional principal.
  - ✓ $\text{BPV} = \$1{,}000{,}000 \times 0.01\% \times 90/360 = \$25$

Summary

- **Importance:** ☆☆
- **Content:**
  - ✓ Futures price vs. forward price;
  - ✓ Interest rate futures.
- **Exam tips:**
  - ✓ 常考点：Forward与Futures价格对比。

高顿教育

Pricing and Valuation of Swap

Tasks:

- Describe how swap contracts are similar to but different from a series of forward contracts;
- Contrast the value and price of swaps.

Pricing and Valuation of Swap

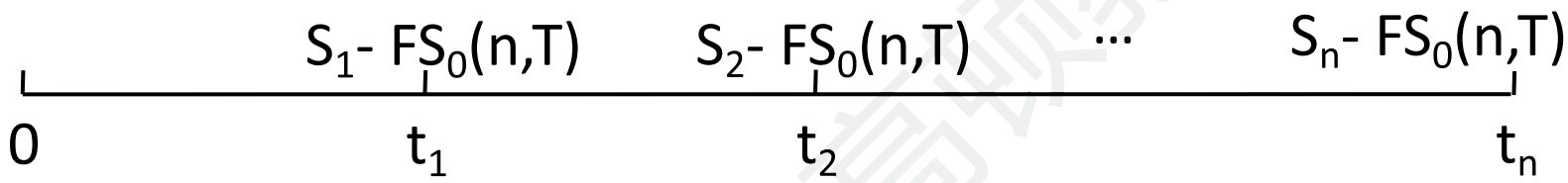
Pricing and valuation of swap

- As with forward contract, the price of a swap is the fixed price specified in the swap contract (the contract price) and the value depends on how expected future price change over time;
  - ✓ At initiation, a swap has zero value;
  - ✓ An increase in expected future price will produce a positive value for the fixed-price payer, and a decrease in expected future price will produce a negative value.

Pricing and Valuation of Swap

Pricing and valuation of swap

- A swap is a series of forwards, each created at swap price.
  - ✓ A swap: make n payments of $FS_0(n, T)$ at time $t_1, t_2, \ldots, t_n$ and receive $S_1, S_2, \ldots, S_n$.



  - ✓ The value of a swap equals the present value of all remaining future swap settlements.

Pricing and Valuation of Swap

Swap vs. forward

- At initiation, the value of each one of the series forwards is not zero.
  - ✓ Off-market forward: forward contract created with a contract price that gives it a non-zero value at initiation.
  - ✓ A swap consist of some off-market forwards with positive present value and some off-market forwards with negative present values, so that the sum of their present values equals zero.

Pricing and Valuation of Swap

Interest rate swap

- In a simple (plain vanilla) interest rate swap, one party pays a floating rate and the other pays a fixed rate on a notional principal amount.
  - ✓ An increase in expected future rate will increase the value for the fixed-rate payer, and vise versa.
- At each payment date, a net payment is made from one party to the other.
  - ✓ $Net\ payment_t = (MRR_{t-1} - Swap\ rate) \times NP \times Period$
    - For fixed rate payer.

Pricing and Valuation of Swap

Interest rate swap (Cont.)

- An interest rate swap is equivalent to a series of (off-market) FRA, each with a forward rate equal to the swap fixed rate.
- The fixed-rate payer is equivalent to being long a floating-rate bond and short a fixed-rate bond.
  - ✓ For fixed-rate payer (floating-rate receiver):

$$V_t = PV_{\text{Floating-rate bond}} - PV_{\text{Fixed-rate bond}}$$

  - ✓ For fixed-rate receiver (floating-rate payer):

$$V_t = PV_{\text{Fixed-rate bond}} - PV_{\text{Floating-rate bond}}$$

Pricing and Valuation of Swap

Interest rate swap (Cont.)

- Due to better liquidity, active fixed-income portfolio managers often use swaps rather than underlying securities to adjust their interest rate exposure.

高顿教育

Summary

- **Importance:** ☆☆
- **Content:**
  - ✓ Pricing and valuation of swap;
  - ✓ Interest rate swap.
- **Exam tips:**
  - ✓ 主要是概念题；
  - ✓ 常考点：无.

高顿教育

Pricing and Valuation of Swap

Interest rate swap

- An interest rate swap is equivalent to a series of (off-market) FRA, each with a forward rate equal to the swap fixed rate.
- Different FRA fixed rates usually exist for different times to maturity.
- Recall: $[1+\mathrm{Z_B}]^{\mathrm{B}} = (1+\mathrm{Z_A})^{\mathrm{A}} \times [1+IFR_{A,B-A}]^{\mathrm{B-A}}$
- In contrast, a standard interest rate swap has a constant fixed rate over its life.

Pricing and Valuation of Swap

Interest rate swap (Cont.)

- The swap rate should equates the present value of all future expected floating cash flows to the present value of fixed cash flows.
  - ✓ $\sum \frac{IFR}{(1+Z_i)^i} = \sum \frac{S_i}{(1+Z_i)^i}$
- The fixed-rate payer is equivalent to being long a floating-rate bond and short a fixed-rate bond.
  - ✓ $\text{PV}_{\text{Floating-rate bond}} = \text{PV}_{\text{Fixed-rate bond}} = \text{Par}$
  - ✓ $\sum \frac{S_i}{(1+Z_i)^i} + \frac{1}{(1+Z_i)^i} = 1$

高顿教育

Pricing and Valuation of Option

Tasks:

- Explain the exercise value, moneyness, and time value of an option;
- Contrast the use of arbitrage and replication concepts in pricing forward commitments and contingent claims;
- Identify the factors that determine the value of an option and describe how each factor affects the value of an option.

Pricing and Valuation of Option

Moneyness

- Moneyness refers to whether an option is in the money or out of the money:
  - ✓ If immediate exercise would generate a positive payoff, the option is in the money (ITM).
  - ✓ If immediate exercise would result in a loss (negative payoff), the option is out of the money (OTM).
  - ✓ If immediate exercise will generate neither a gain nor loss, the option is at the money (ATM).

Pricing and Valuation of Option

Moneyness (X: exercise price, S: spot price)

- Call option
  - ✓ In the money: S > X;
  - ✓ Out of the money: S < X;
  - ✓ At the money: S = X.
- Put option
  - ✓ In the money: S < X;
  - ✓ Out of the money: S > X;
  - ✓ At the money: S = X.

高顿教育

Pricing and Valuation of Option

Exercise value (intrinsic value)

- The option contract's value if the option were exercisable at time t.
- The exercise value at time t incorporating the time value of money is the difference between the spot price (St) and the present value of the exercise price (PV(X)).
  - ✓ Call option: $\mathrm{Max}[0, S_t - \mathrm{X}(1+r)^{-(T-t)}]$
  - ✓ Put option: $\mathrm{Max}[0, \mathrm{X}(1+r)^{-(T-t)} - S_t]$

Pricing and Valuation of Option

Time value (speculative value)

- The amount by which the current option premium (price) exceeds the intrinsic value.
  - ✓ Option premium = Intrinsic value + Time value
- A longer time until expiration usually means a higher potential dispersion of the future underlying price.
- Is always positive but declines to zero at maturity, a process referred to as **time value decay**.

Pricing and Valuation of Option

Artibrage

- Conditional nature of option payoff profiles leads us to establish upper and lower no-arbitrage price bounds at any time t.
- For call option:
  - ✓ $\mathrm{Max}[0, S_t - \mathrm{X}(1+r)^{-(T-t)}] < C_t \leq S_t$
- For put option:
  - ✓ $\mathrm{Max}[0, \mathrm{X}(1+r)^{-(T-t)} - S_t] < P_t \leq X$

高顿教育

Pricing and Valuation of Option

Replication

- Recall: Asset - Risk-free asset = Derivative
  - ✓ Long forward = long underlying + short risk free asset
  - ✓ Short forward = long risk free asset + short underlying
- For option, exercise are uncertain.
- Option replicating transaction should be adjusted for probability of exercise.

Factors that Determine Option Value

Price of the underlying asset (S_t)

- For call option, a higher S_t will increase its value;
- For put option, a higher S_t will decrease its value.

The exercise price (X)

- For call option, a higher X will decrease its value;
- For put option, a higher X will increase its value.

Factors that Determine Option Value

The risk-free rate of interest (R_f)

- For call option, a higher R_f will increase its value;
- For put option, a higher R_f will decrease its value.

Volatility of the underlying price (δ)

- A higher δ will increase the value of both call and put options.

Factors that Determine Option Value

Time to expiration (T-t)

- For call option and most put option, a longer (T-t) will increase the time value and then increase its value;
- For some European put options, a longer (T-t) are more likely to decrease its value when:
  - ✓ The deeper a put option is in the money;
  - ✓ The higher the risk-free rate;
  - ✓ The longer the current time to expiration.

Factors that Determine Option Value

Costs (c) and benefits (i) of holding the asset

- For call option, a higher cost (c) will increase its value and a higher benefit (i) will decrease its value;
- For put option, a higher cost (c) will decrease its value and a higher benefit (i) will increase its value.
  - ✓ Reason: cost will generally increase the price of the underlying asset, and benefit will generally decrease the price of the underlying asset.

Factors that Determine Option Value

Summary of factors determining option value

	Call option	Put option
S	↑	↓
X	↓	↑
R_f	↑	↓
δ	↑	↑
T-t	↑	↑ (with exception)
c	↑	↓
i	↓	↑

- Note: ↑ means positively related, and ↓means negatively related.

Summary

- **Importance:** ☆☆☆
- **Content:**
  - ✓ Option value: intrinsic value + time value;
  - ✓ Arbitrage and replication for option;
  - ✓ Factors on option value: S, X, Rf, δ, T-t, c, i.
- **Exam tips:**
  - ✓ 常考点1：影响option value的factors，及影响方式；

    Intrinsic value与time value的理解及计算。

Option Replication Using Put–Call Parity

Tasks:

- Explain put–call parity for European options;
- Explain put–call forward parity for European options.

Put-Call Parity

Fiduciary call $(\mathbf{c} + \mathbf{X}/(\mathbf{1} + \mathbf{R_f})^{\mathbf{T}})$

- Combination of a European call option with exercise price of X and a pure-discount, riskless bond with face value of X.

Protective put $(\mathbf{S} + \mathbf{p})$

- A share of stock together with a put option on the stock with exercise price of X.
- **Note:** the call and put must be European-style (c, p) with same exercise price (X) and time to expiration (T). Also, the maturity of the bond should be the same (T).

Put-Call Parity

Payoff at expiration:

	Fiduciary call $(c + X/(1+R_f)^T)$	**Protective put** $(S + p)$
If $S \leq X$	$0 + X = X$	$S + (X - S) = X$
If $S \geq X$	$(S - X) + X = S$	$S + 0 = S$

- In either case, the payoff on a fiduciary call is the same as the payoff on a protective put, so:
  - ✓ Put-call parity: $S_0 + p_0 = c_0 + X/(1 + R_f)^T$

Put-Call Parity

Arbitrage opportunity

- If $S_0 + p_0 > c_0 + \mathrm{X}/(1+\mathrm{R_f})^{\mathrm{T}}$
  - ✓ Selling the put and the underlying and purchasing the call and the risk-free asset at initiation.
- If $S_0 + p_0 < c_0 + \mathrm{X}/(1+\mathrm{R_f})^{\mathrm{T}}$
  - ✓ Purchasing the put and the underlying and selling the call and the risk-free asset at initiation.

Put-Call Parity

Replication

- The put-call parity can be rearranged to create synthetic equivalencies.
  - ✓ Synthetic call: $c = S + p - X/(1 + R_f)^T$
  - ✓ Synthetic put: $p = c + X/(1 + R_f)^T - S$
  - ✓ Synthetic stock: $S = c + X/(1 + R_f)^T - p$
  - ✓ Synthetic bond: $X/(1 + R_f)^T = S + p - c$
  - ✓ **Note:** the term "+" means long and "-" means short.

Put-Call Parity

Put-call forward parity

- Put-call-forward parity is derived with a forward contract (forward price: $F_0(T)$) rather than the underlying asset itself.
  - ✓ Substituting $F_0(T)/(1+R_f)^T$ for S_0 in put-call parity gives us the put-call-forward parity:
    - $F_0(T)/(1+R_f)^T + p = c + X/(1+R_f)^T$
    - Or:
    - $[F_0(T) - X]/(1+R_f)^T = c - p$

Summary

- **Importance:** ☆☆☆
- **Content:**
  - ✓ Put-call parity;
  - ✓ Put-call forward parity.
- **Exam tips:**
  - ✓ 常考点1：Put-call parity计算；
    Put-call parity复制。

高顿教育

Put-Call Parity

Put-call parity application

- Put-call parity can be used to model the value of a firm's equity holder and debt holder.
  - ✓ The company defaults when the value of its assets falls below the default barrier (i.e., the liability).
  - ✓ It assumes limited liability and equity values are non-negative.

Put-Call Parity

Put-call parity application (Cont.)

- **Equity option analogy**: holding the company's equity is economically equivalent to owning a **European call option** on the company's assets (A) with strike price K and maturity T (maturity of debt).
- At maturity:
  - ✓ If A > K (Solvency), exercise the call option and realize a value of (A-K).
  - ✓ If A < K (Insolvency), do not exercise the call option and the value is 0.

Put-Call Parity

Put-call parity application (Cont.)

- **Debt option analogy**: owning the company's debt is economically equivalent to owning **a riskless bond**, and simultaneously **selling a European put option** on the assets (A) with strike price K and maturity T (maturity of debt).

$$V_{\text{risky debt}} = V_{\text{risk-free debt}} - V_{\text{put option}}$$

- At maturity:
  - ✓ If A < K (Solvency), exercise the put option and the value is A.
  - ✓ If A > K (Insolvency), do not exercise the put option and the value is K.

Valuing a Derivative Using Binomial Models

Tasks:

- Explain how to value a derivative using a one-period binomial model;
- Describe the concept of risk neutrality in derivatives pricing.

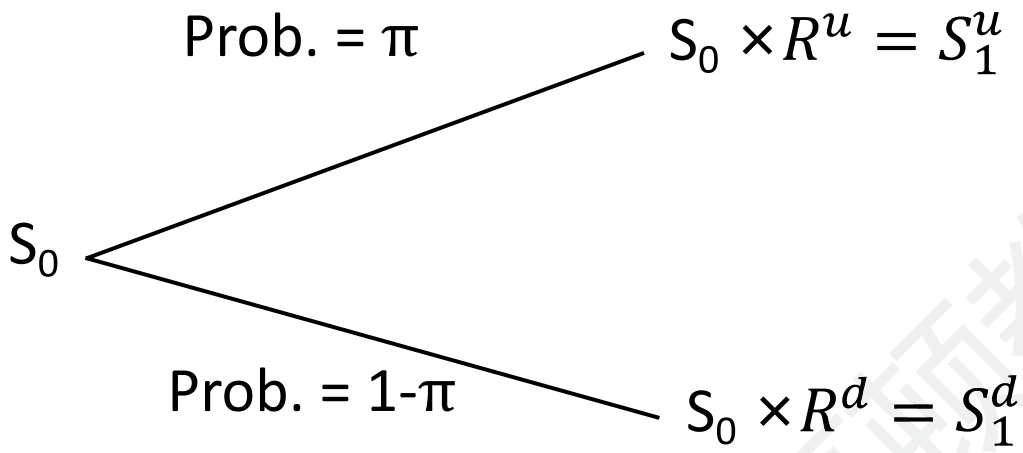
Binomial Model

One-period binomial model

- Binomial model is based on the idea that, over the next period, some value will change to one of two possible values.
- To construct a binomial model, we need to know the beginning asset value (S_0), the size of the two possible changes (R^u, R^d), and the probabilities of each of these changes occurring (π, $1-\pi$).
  - ✓ $R^u = S_1^u / S_0 > 1$
  - ✓ $R^d = S_1^d / S_0 < 1$

Binomial Model

One-period binomial model (Cont.)



- ✓ $\pi = (1 + R_f - R^d)/(R^u - R^d)$, risk-neutral probability of an up-move;
- ✓ $1 - \pi$, risk-neutral probability of an down-move.

Binomial Model

One-period binomial model (Cont.)

- With one-period binomial model, the value of an option on stock can be calculated as:
  - ✓ Step 1: Calculate the payoff of the option at maturity in both the up-move and down-move states;
  - ✓ Step 2: Calculate the expected value of the option in one period as the probability-weighted average of the payoffs in each state;
  - ✓ Steps 3: Discount this expected value back to today at the risk-free rate.

Binomial Model

Example

- A non-dividend-paying stock is currently trading at €80. A put option on the stock has one year to mature and exercise price of €80. Assume the risk-free interest rate is 5% and a single-period binomial option valuation model where R^u = 1.375 and R^d = 0.75, calculate the put option value.

Binomial Model

Answer

- $S_1^u = 1.375 \times 80 = 110; S_1^d = 0.75 \times 80 = 60$
- $p_1^u = \text{Max}(0, \text{X} - S_1^u) = 0; p_1^d = \text{Max}(0, \text{X} - S_1^d) = 20$
- $\pi = (1 + \text{R}_\text{f} - R^d)/(R^u - R^d) = (1 + 5\% - 0.75)/(1.375 - 0.75) = 0.48$
- Value of the put option:

$$p_0 = [\pi \times p_1^u + (1 - \pi) \times p_1^d]/(1 + \text{R}_\text{f})^\text{T}$$

$$= \frac{0.48 \times 0 + 0.52 \times 20}{1 + 5\%} = 9.90$$

Binomial Model

Risk neutrality

- An option's value is not affected by actual (real-world) probabilities of underlying price increases or decreases.
  - ✓ The risk-neutral probabilities are determined solely by the up and down gross returns.
- This no-arbitrage derivative value established separately from investor views on risk is referred to as risk-neutral pricing.

Binomial Model

Option hedge ratio

- The proportion of the underlying that will offset the risk associated with an option.
- For call option:
  - ✓ $h_c = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$ short h_c underlying to hedge a call
- For put option:
  - ✓ $h_p = \frac{p_1^u - p_1^d}{S_1^u - S_1^d}$ short h_p underlying to hedge a put

Summary

- **Importance:** ☆☆☆
- **Content:**
  - ✓ One-period binomial model;
  - ✓ Risk neturality;
  - ✓ Hedge ratio.
- **Exam tips:**
  - ✓ 常考点1：One-period binomial model中的概念。

高顿教育

Binomial Model

One-period binomial model

- Hedged portfolio: long an option + short h underlying.
  - ✓ $V_0 = c_0 - h \times S_0$
- The hedged portfolio should have the same value in up-move scenario and down-move scenario.
  - ✓ $V_1^u = c_1^u - h \times S_1^u$
  - ✓ $V_1^d = c_1^d - h \times S_1^d$
- The expected return is risk-free rate.
  - ✓ $V_1^d = V_1^u = V_0(1 + r)$

Binomial Model

Example

- A non-dividend-paying stock is currently trading at €80. A call option on the stock has one year to mature and exercise price of €80. Assume the risk-free interest rate is 5% and a single-period binomial option valuation model where R^u = 1.375 and R^d = 0.75, calculate the call option value.

Binomial Model

Answer (Method 1)

- $S_1^u = 1.375 \times 80 = 110; S_1^d = 0.75 \times 80 = 60$
- $c_1^u = \text{Max}(0, S_1^u - X) = 30; c_1^d = \text{Max}(0, S_1^d - X) = 0$
- $\pi = (1 + \text{R}_\text{f} - R^d)/(R^u - R^d) = (1 + 5\% - 0.75)/(1.375 - 0.75) = 0.48$
- Value of the call option:

$$p_0 = [\pi \times c_1^u + (1 - \pi) \times c_1^d]/(1 + \text{R}_\text{f})^\text{T}$$

$$= \frac{0.48 \times 30 + 0.52 \times 0}{1 + 5\%} = 13.71$$

Binomial Model

Answer (Method 2)

- $V_1^u = c_1^u - h \times S_1^u = V_1^d = c_1^d - h \times S_1^d$
  - ✓ $30 - \mathrm{h} \times 110 = 0 - \mathrm{h} \times 60 \rightarrow \mathrm{h} = 0.6$
  - ✓ $V_1^u = V_1^d = 30 - 0.6 \times 110 = -36$
- $V_0 = c_0 - h \times S_0 = V_1^d/(1+r)$
  - ✓ $c_0 - 0.6 \times 80 = -36/(1+5\%) \rightarrow c_0 = 13.71$
- The same as method 1.